



A deep learning based multi-model approach for predicting drug-like chemical compound's toxicity

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ABSTRACT

Ensuring the safety and efficacy of chemical compounds is crucial in small-molecule drug development. In the later stages of drug development, toxic compounds pose a significant challenge, losing valuable resources and time. Early and accurate prediction of compound toxicity using deep learning models offers a promising solution to mitigate these risks during drug discovery. In this study, we present the development of several deep-learning models aimed at evaluating different types of compound toxicity, including acute toxicity, carcinogenicity, hERG cardiotoxicity (the human ether-a-go-go related gene caused cardiotoxicity), hepatotoxicity, and mutagenicity. To address the inherent variations in data size, label type, and distribution across different types of toxicity, we employed diverse training strategies. Our first approach involved utilizing a graph convolutional network (GCN) regression model to predict acute toxicity, which achieved notable performance with Pearson R 0.76, 0.74, and 0.65 for intraperitoneal, intravenous, and oral administration routes, respectively. Furthermore, we trained multiple GCN binary classification models, each tailored to a specific type of toxicity. These models exhibited high area under the curve (AUC) scores, with an impressive AUC of 0.69, 0.77, 0.88, and 0.79 for predicting carcinogenicity, hERG cardiotoxicity, mutagenicity, and hepatotoxicity, respectively. Additionally, we have used the approved drug dataset to determine the appropriate threshold value for the prediction score in model usage. We integrated these models into a virtual screening pipeline to assess their effectiveness in identifying potential low-toxicity drug candidates. Our findings indicate that this deep learning approach has the potential to significantly reduce the cost and risk associated with drug development by expediting the selection of compounds with low toxicity profiles. Therefore, the models developed in this study hold promise as critical tools for early drug candidate screening and selection.

1. Introduction

The safety and efficiency of drug-like chemical compounds are pivotal factors in small-molecule drug development [1,2]. Toxicity-related issues frequently hinder progress, leading to the termination of potential drug candidates [3,4]. These setbacks result in substantial financial losses and impede the delivery of safe and effective therapies to needy patients [5]. To address this challenge, early identification of compound toxicity is crucial [6]. Numerous types of drug toxicity exist, each with specific concerns and implications [7]. It is imperative to

consider the most frequent and essential drug toxicity to develop effective prediction models [8]. Adverse drug reactions (ADRs) are a significant concern and encompass a wide range of toxicities, such as hepatotoxicity (liver toxicity), cardiotoxicity (heart toxicity), nephrotoxicity (kidney toxicity), and neurotoxicity (nervous system toxicity) [9]. The toxicity accounts for significant drug failures and can severely affect patient safety and well-being [3]. However, traditional methods, such as *in vitro* or *in vivo* experiments to evaluate drug toxicity, are time-consuming and costly [10]. Despite this, toxicity studies can be biased due to species variation or lack of appropriate animal research models

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Table 1

The data size of the training and test set for toxicity prediction models.

Model type	Data set				Train size		Test size
Regression	Acute Toxicity mouse intraperitoneal				32,000		3,079
	Acute Toxicity mouse intravenous				14,000		2,248
	Acute Toxicity mouse oral				20,000		2,512
Model type	Data set	Train size	Test size	Train positive size	Train negative size	Test positive size	Test negative size
Binary classification	Carcinogenicity	800	200	397	403	103	97
	Cardiotoxicity	1,000	295	456	544	134	161
	Hepatotoxicity	2,500	303	1,281	1,219	162	141
	Mutagenicity	5,500	1,376	3,034	2,466	756	620

[11], which may miss some potential adverse effects, thereby exposing patients or volunteers to a relatively high risk [6].

Several computational methods exist for evaluating drug toxicity, including quantitative structure–activity relationships (QSAR), molecular docking, and machine learning-based models [12–15]. QSAR models have been used extensively in predicting drug toxicity, relying on statistical models to relate molecular features to toxicity [16,17]. However, QSAR usually needs to spend time extracting feature-based experience, and those extracted features may lose some important toxicity-related features [18]. Molecular docking calculates the binding energy between a drug and its target molecule, providing insight into the molecular mechanism of action and possible sites of toxicity [19,20]. However, molecular docking usually can only evaluate off-target effects, and it is very time-consuming to evaluate large-scale compounds because each compound needs inverse docking with many toxicities related to protein [21]. It should be noted that the accuracy of docking is not reliable.

Deep learning-based approaches have shown significant potential in compound toxicity prediction in recent years [22–25]. Particularly in the early stage of drug screening, computational methods offer several advantages over traditional methods, including speed, cost-effectiveness, and the ability to evaluate large numbers of compounds simultaneously, leading to a significant improvement in the success rate of drug development [26–30]. Due to these reasons, many important drug regulatory agencies, including the Food and Drug Administration (FDA) and European Medicines Agency (EMA), have recognized the potential of Artificial Intelligence (AI) in drug toxicity prediction and are closely monitoring its development [31–33]. However, the prediction of different types of compound toxicity necessitates the use of specialized models due to variations in data characteristics [34,35]. Deep learning models can potentially learn complex, nonlinear relationships between molecular features and toxicity [36,37]. They can also accommodate molecule's diverse and complex structures [38].

In this study, we aimed to develop multiple deep-learning models that effectively evaluate compound toxicity, including acute toxicity, carcinogenicity, hERG_cardiotoxicity, and mutagenicity. The primary objective was to enhance the predictive accuracy for each toxicity type while addressing the challenges associated with varying dataset sizes, label types, and distributions. Evaluating drug toxicity is one of the most critical steps in the drug development process. Patients' safety and avoiding side effects are paramount to pharmaceutical companies. Our models have the potential to significantly improve the efficiency, cost-effectiveness, and accuracy of predicting drug toxicity in the early stages of drug discovery, reduce the frequency of non-clinical and clinical trial failures, and minimize the risks associated with adverse effects. Furthermore, our models could be integrated into the drug development pipeline, supporting optimization and selecting safer and more effective compounds.

2. Method

2.1. Data collection

The toxicity data utilized in this study were obtained from the TOXRIC online database [39] (<https://toxric.bioinformai.tech/download>). Four compound toxicities were selected: acute toxicity, carcinogenicity, hERG_cardiotoxicity, and mutagenicity. Two specific toxicity parameters were considered for acute toxicity: mouse_intraperitoneal LD50 and mouse_intravenous LD50. These parameters represent mice's Lethal Dose Fifty (LD50) values via the intraperitoneal and intravenous routes, respectively. Two independent models, model_AT_X and model_AT_Y, were trained using these datasets to predict toxicity based on the respective administration routes. We utilized the carcinogenicity dataset available in the TOXRIC database to predict carcinogenicity. This dataset comprises data from the Carcinogenic Potency Database (CPDB) summary tables (CPDBAS, version 5d). The CPDB offers standardized and comprehensive information on the long-term results of animal carcinogenesis studies [40].

For the hERG_cardiotoxicity, cardiotoxicity-5 data was used; this file contains the data about undesirable hERG-related cardiotoxicity of drugs. This file sets IC50 = 5 as the threshold to discriminate hERG blockers from nonblockers. Compared to other thresholds, the positive and negative data are relatively balanced.

For hepatotoxicity, the LTKB (Liver Toxicity Knowledge Base) database was used; LTKB is a project at the FDA's National Center for Toxicological Research to study drug-induced liver injury (DILI) [41].

For Mutagenicity prediction, we employed the Ames Mutagenicity dataset from TOXRIC. This dataset includes compounds with known mutagenic properties, encompassing 4205 mutagens and 3312 non-mutagens [39].

2.2. Training and testing dataset

We have prepared the input data for training and testing for those different types of toxicity, shown in Table 1. For the model of Acute-Toxicity_mouse_intraperitoneal, the training data size is 32,000, and the testing size is 3,079. For the model of Acute-Toxicity_mouse_intravenous, the training and testing data sizes are 14,000 and 2,248, respectively. The training and testing data sizes for carcinogenicity are 800 and 200, respectively. The training data contains 397 and 403 samples for positive and negative, respectively. The testing set includes 103 and 97 samples for positive and negative, respectively.

For the cardiotoxicity, the training and testing data sizes are 1,000 and 295, respectively. The training data contains 456 and 544 samples for positive and negative, respectively. The testing set includes 134 and 161 samples for positive and negative, respectively.

The training and testing data sizes for hepatotoxicity are 2,500 and 303, respectively. The training data contains 1,281 and 1,219 samples for positive and negative, respectively. The testing set includes 162 and 141 samples for positive and negative, respectively.

The training and testing data for the mutagenicity are 5,500 and

Table 2

The sizes of the extra test datasets.

Model	Extra test set	Size
Model_AT_X	Acute Toxicity rat intraperitoneal	2,173
Model_AT_Y	Acute Toxicity rat intravenous	865
Model_AT_Z	Acute Toxicity rat oral	5,198

1,376, respectively. The training data contains 3,034 and 2,466 positive and negative samples, respectively. The testing set includes 756 and 620 samples for positive and negative, respectively.

2.3. An extra test set for acute toxicity prediction

We obtained three rat-related datasets (Acute Toxicity rat intraperitoneal, Acute Toxicity rat intravenous, and Acute Toxicity rat oral) from the TOXRIC online database. We excluded those compounds that have occurred in the corresponding mouse-related dataset; in such a way, the compounds do not occur at each training set of the models. The dataset size is shown in Table 2.

2.4. Small molecule representation

We used the graph to represent the small molecule, with the atom as

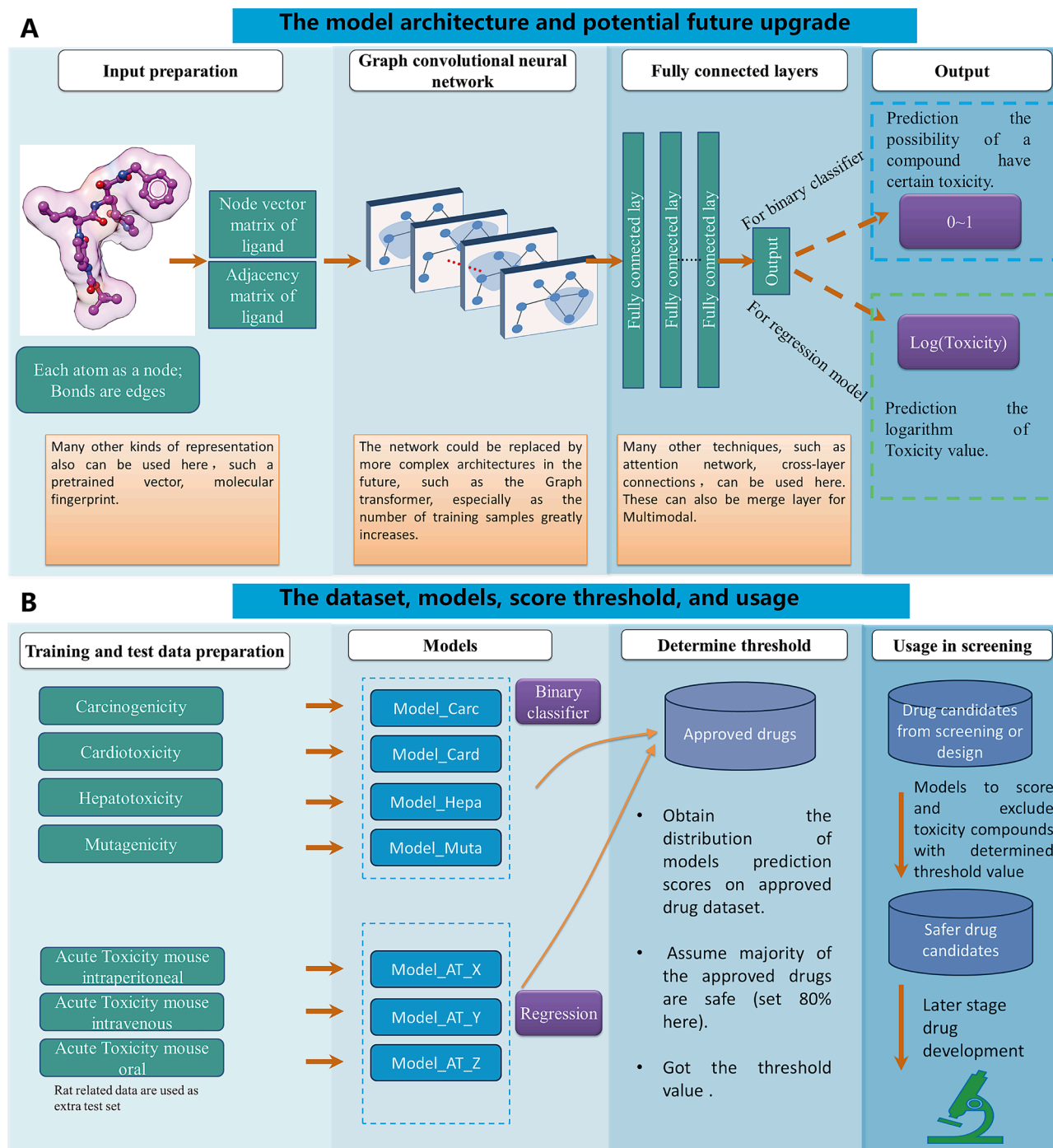


Fig. 1. Illustration of model structure, training, and usage. A. Model input representation, Neural Network architecture, meaning of model output, and potential future upgrade Strategy; B. Datasets collection, models obtained, threshold determination strategy, and potential applications in real drug screening or design.

a node and the bond as an edge [42].

2.5. Model construction

Our model structure is depicted in Fig. 1A, which uses the molecular graph as input.

The input features are passed through four GCNConv layers sequentially. The first layer, GCNConv, maps the input features of dimension 'num_features'(78) to the same dimensional space. The second GCNConv layer doubles the feature dimensions. The third GCNConv layer increases the features from 'num_features*2' to 'num_features*4'. The fourth and final GCNConv layer keeps the dimension consistent as 'num_features*4'.

After the GCNConv layers, a linear transformation is applied where the input dimensions are reduced from 'num_features*4' to 1,024 with a linear layer denoted as 'fc_g1'. A subsequent linear layer 'fc_g2' maps this 1,024-dimensional input to an 'output_dim'(1280)-dimensional output.

A Rectified Linear Unit (ReLU) activation function is applied after the linear transformations, introducing non-linearity to the model. A dropout layer is also included to prevent overfitting, with a specified dropout rate of 0.2.

Following this, there are three more fully connected layers. The first two, 'fc1' and 'fc2', perform transformations from 'output_dim' to 1,024 dimensions and then to 512 dimensions, successively. The final linear layer 'out' reduces the dimensionality from 512 to a single output node. We used the linear activation function to output the node for regression prediction, which returns a real number. We use Adam as the optimizer and MSELoss (Mean Squared Error Loss) as the loss function in a regression model. We used the sigmoid activation function to the output node for the binary classification model, which returns a value between 0 and 1. We use Adam as an optimizer and BCELoss (Binary Cross Entropy Loss) as a loss function in the binary classification model. Other parameters are as follows:

LR (learning rate) = 0.0005

NUM_EPOCHS (number of epochs) = 2,000

TRAIN_BATCH_SIZE (training batch size) = 2,000

Overall, this structure incorporates both the strength of Graph Convolutional Networks for effectively extracting features in graph-structured data and the power of deep feedforward networks in toxicity prediction-specific feature learning.

2.6. Model training

The torch_geometric module created input data and constructed the graph neural network. The input data was saved in PyTorch InMemoryDataset format. The PyTorch was used to do the training. We chose BCELoss (Binary Cross Entropy Loss) as the loss function and adopted the Adam optimizer. The learning rate was 0.0005, and the total number of epochs was set to 2,000. We save the model every 100 epochs. The number of epochs that we finally chose was based on the performance convergences on the test set. We recorded performance on the validation set every epoch to observe the model convergences and conduct different epoch-sensitive analyses.

2.7. Evaluation metrics

2.7.1. Performance metrics for the regression model

We used RMSE (Root Mean Squared Error), MSE (Mean Squared Error), Pearson (Pearson correlation coefficient), Spearman (Spearman correlation coefficient), and C-index (Concordance Index, CI) to evaluate the performance of the regression model. RMSE calculates the square root of the average of the squared differences between the

predicted values and the actual values. MSE is the average of the squared differences between the predicted and actual values. Pearson correlation measures the linear relationship between the predicted and actual values. Spearman correlation is a non-parametric measure that assesses the monotonic relationship between variables. C-index measures the ability of a model to rank the observed outcomes in terms of their relative risk or event occurrence probabilities.

2.7.2. Performance metrics for binary classification model

We have used AUC (Area Under the ROC Curve), TPR (True Positive Rate), Precision, Accuracy, and MCC (Matthews Correlation Coefficient) as evaluation metrics for the model performance. AUC represents the area under the ROC (Receiver Operating Characteristic) curve. ROC curve is used to evaluate the performance of a binary classification model by plotting the true positive rate (TPR) against the false positive rate (FPR) at different threshold values. TPR is the ratio of true positives to the sum of true positives and false negatives. TPR represents the proportion of actual positives that the model correctly classifies. TPR is calculated using the following formula:

$$TPR = TP / (TP + FN)$$

where TP represents the number of true positive samples, and FN represents the number of false negative samples.

2.7.3. Precision

Precision is the ratio of true positives to the sum of true and false positives. Precision represents the proportion of true positives to the total predicted positives by the model. Precision is calculated using the following formula:

$$Precision = TP / (TP + FP)$$

where TP represents the number of true positive samples, and FP represents the number of false positive samples.

2.7.4. Accuracy

Accuracy is the ratio of correctly classified samples to the total number of samples. The value of accuracy ranges from 0 to 1.

Accuracy is calculated using the following formula:

$$Accuracy = (TP + TN) / (TP + TN + FP + FN)$$

where TP represents the number of true positive samples, TN represents the number of true negative samples, FP represents the number of false positive samples, and FN represents the number of false negative samples.

2.7.5. MCC (Matthews correlation coefficient)

MCC is a correlation coefficient between the actual and predicted binary classifications. It takes values between -1 to 1. A value of -1 indicates a perfect negative correlation, 1 indicates a perfect positive correlation, and 0 indicates no correlation. MCC is widely used to evaluate the performance of binary classification models, especially when the dataset is imbalanced.

MCC is calculated using the following formula:

$$MCC = (TP * TN - FP * FN) / \sqrt{(TP + FP) * (TP + FN) * (TN + FP) * (TN + FN)}$$

where TP represents the number of true positive samples, TN represents the number of true negative samples, FP represents the number of false positive samples, and FN represents the number of false negative samples.

2.8. Data split strategy

All the datasets were divided using a random split strategy. In this

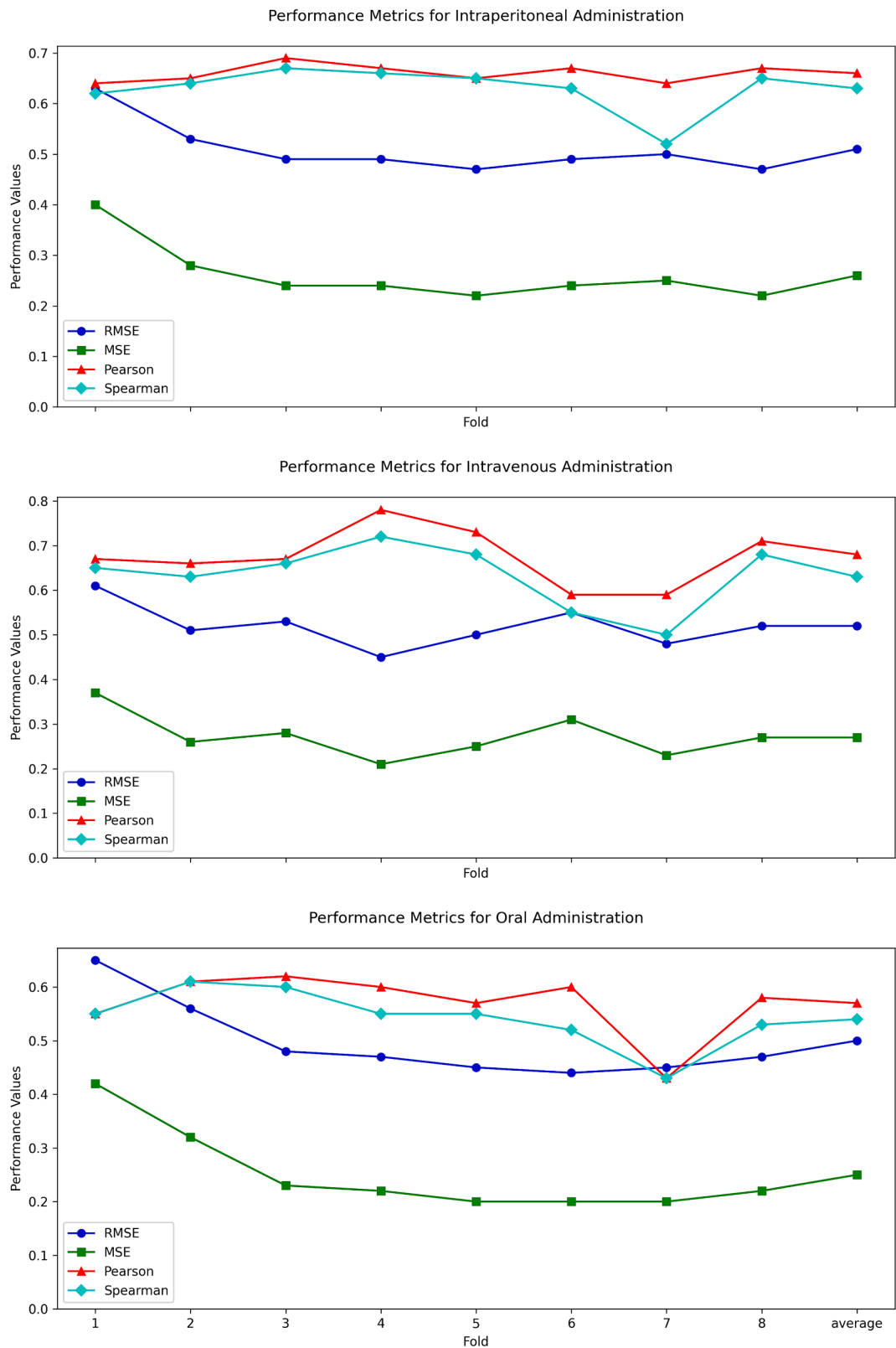


Fig. 2. The performance of 8-fold cross-validation for models trained over intraperitoneal intravenous and oral, respectively.

strategy, the original dataset is randomly split into two parts: a training set and a testing set; detailed training and testing set sizes are shown in Table 1. We also used 8-fold cross-validation (CV) for the acute toxicity. During 8-fold CV, the original dataset is randomly divided into eight equal-sized subsets called “folds”. Subsequently, the model is trained on

seven folds while the remaining fold is used for testing. This iterative process is repeated eight times, ensuring that each fold serves as the testing set once while the other seven folds are employed for training.

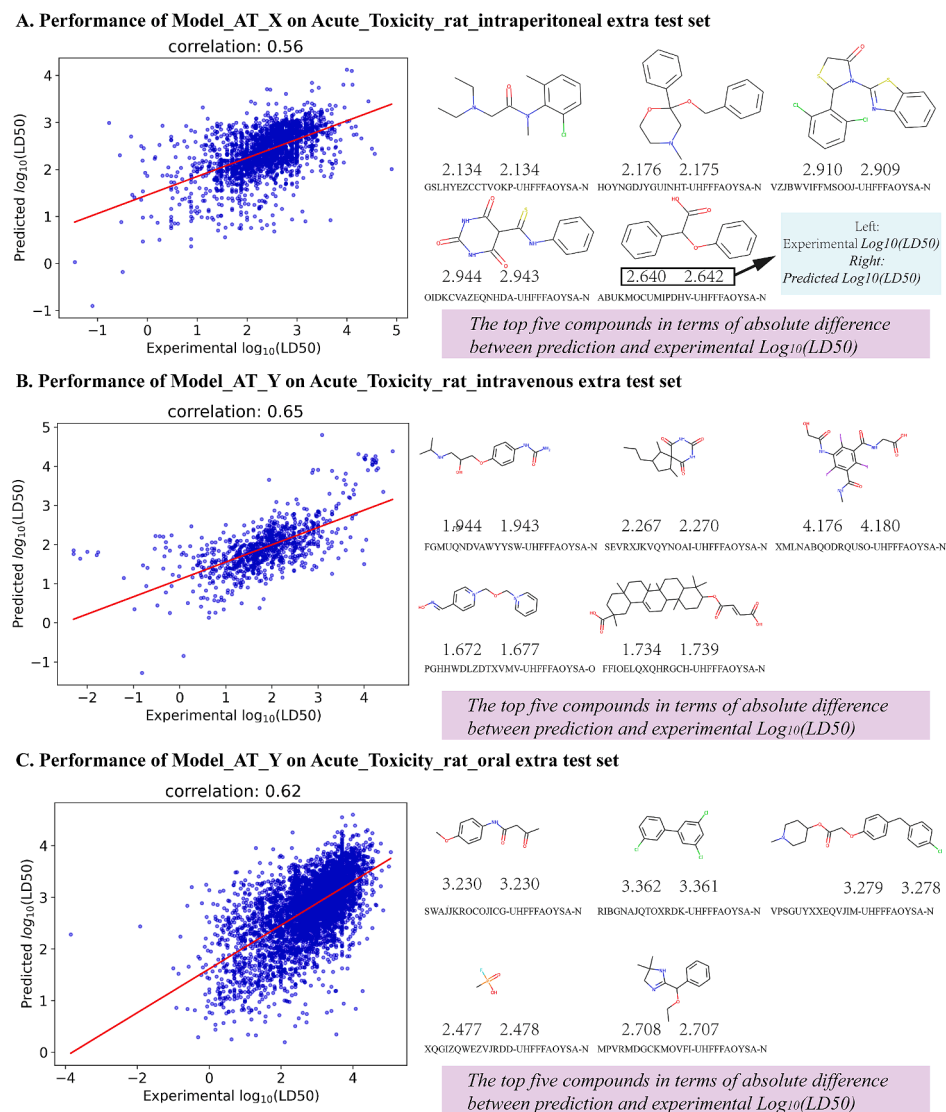


Fig. 3. The performance of three acute toxicity models on an extra test set. A, the performance of intraperitoneal related model AT_X on the rat intraperitoneal data, the top five best-predicted compounds in terms of the absolute difference between prediction and experimental $\log_{10}(\text{LD50})$ are shown in the right panel; B, the performance of intraperitoneal related model AT_Y on the rat intraperitoneal data, the top five best-predicted compounds in terms of the absolute difference between prediction and experimental $\log_{10}(\text{LD50})$ are shown in right panel; C, the performance of intraperitoneal related model AT_Z on the rat intraperitoneal data, the top five best-predicted compounds in terms of absolute difference between prediction and experimental $\log_{10}(\text{LD50})$ are shown in right panel.

3. Results

In this work, we have built several models to predict important and common compound toxicity that often requires special attention during drug development. The toxicity we focused on in this work is acute toxicity, carcinogenicity, hERG_cardiotoxicity, and mutagenicity, as shown in Fig. 1B. We have developed three models for acute toxicity based on intraperitoneal, intravenous, and oral data, respectively. We choose a regression model for acute toxicity prediction and binary classification models with an output range of 0 ~ 1 for carcinogenicity, hERG_cardiotoxicity, and mutagenicity predictions. The overall workflow of this work is shown in Fig. 1. The developed models have significant potential for widespread use in selecting lower toxicity candidates during drug screening or design processes, thereby significantly reducing costs in late-stage drug development. Additionally, we have proposed future strategies for upgrading the models that can be implemented once the dataset size greatly increases. To enhance the reliability of the application, we have devised an efficient approach to determine the proper threshold value for scoring. It is worth noting that

other types of toxicity, such as reproductive toxicity, are also of great importance; however, the availability of toxicity-related datasets for these types is currently limited. Once datasets for other types of toxicity accumulate, our strategy can be utilized to build specific prediction models for them.

3.1. Model performance of acute toxicity on test set

The three acute toxicity models converge after around 400 epochs, as shown in Fig. S2. We can observe that the performance metrics RMSE, MSE, Pearson correlation, Spearman correlation, and CI in Fig. S2 all have relatively stable values after 400 epochs. The performance of acute toxicity prediction models with different epochs is shown in Table S1. The intraperitoneal-based model has performed well on the test set, with 0.44, 0.2, 0.76, 0.76, and 0.79 for RMSE, MSE, Pearson, Spearman, and C-index (CI) at epoch 2,000, respectively. The intravenous-based model demonstrates promising predicting power on the test set, with 0.49, 0.24, 0.74, 0.71, and 0.77 for RMSE, MSE, Pearson, Spearman, and C-index (CI) at epoch 2,000, respectively. The oral-based model

Table 3
The performance of carcinogenicity, hERG_cardiotoxicity, mutagenicity, and hepatotoxicity model at 2000 epoch.

Model	AUC	TPR	Precision	Accuracy	MCC
Carcinogenicity	0.69	0.66	0.64	0.64	0.27
hERG_cardiotoxicity	0.77	0.69	0.68	0.71	0.42
Mutagenicity	0.88	0.83	0.83	0.81	0.62
Hepatotoxicity	0.79	0.73	0.75	0.73	0.46

demonstrates predicting power on the test set, with 0.46, 0.21, 0.65, 0.66, and 0.75 for RMSE, MSE, Pearson, Spearman, and C-index (CI) at epoch 2,000, respectively. To ensure that the model provides sustained performance, we performed 8-fold cross-validation for models over intraperitoneal intravenous and oral, respectively. In an 8-fold CV, the original dataset is randomly partitioned into eight equally sized subsets, or “folds.” The model is then trained on seven folds and tested on the remaining fold. This process is repeated eight times, each of the eight folds serving as the testing set once, while the other seven folds are used for training. The 8-fold cross-validation (CV) results at the 2,000 epoch

are shown in Fig. 2. The average of the eight test results is typically used as the final performance metric for the model. We can observe that the eight intraperitoneal-based test results achieve an average of 0.51, 0.26, 0.66, and 0.63 for RMSE, MSE, Pearson, and Spearman, respectively. The intravenous-based model averages 0.52, 0.27, 0.68, and 0.63 for RMSE, MSE, Pearson, and Spearman, respectively. The oral-based model averages 0.50, 0.25, 0.57, and 0.54 for RMSE, MSE, Pearson, and Spearman, respectively. The results indicate our model can provide relatively sustained performance.

3.2. Models’ performance of acute toxicity on the extra testing set

The selected test set consists of three data sets regarding acute toxicity in rats, shown in Table 2. These data sets represent the effects of intraperitoneal, intravenous, and oral administration; since they are based on rats, the data is relatively dependent on the training dataset. It is worth noting that mice are considered to be similar to rats in this context. The performance of the models used in the study achieved Pearson correlation coefficients of 0.56, 0.65, and 0.62 for the respective administration routes, as illustrated in Fig. 3. We can observe that

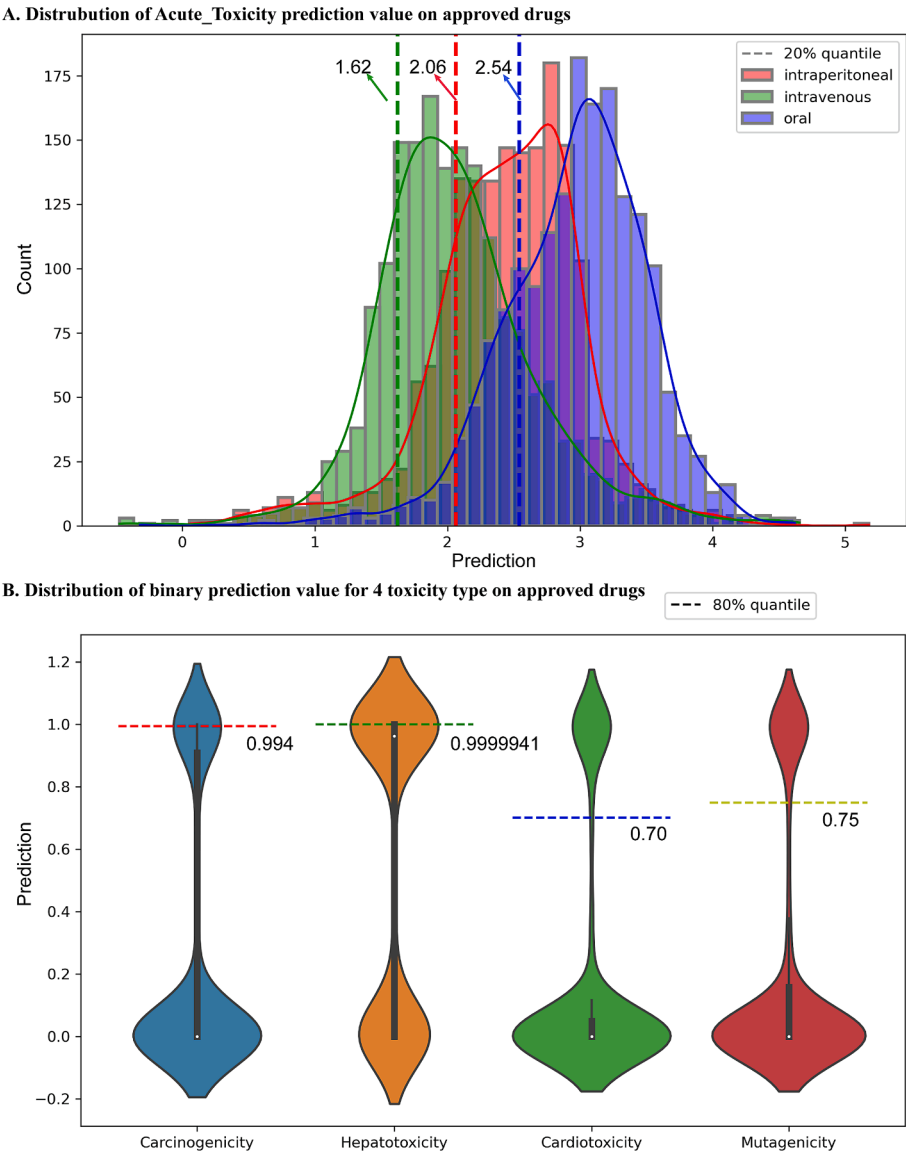


Fig. 4. The distribution of different toxicity models predicting score over the approved drug dataset. A. Score distribution of the regression prediction of models over approved drugs for acute toxicity. B. Score distribution of the binary classification prediction of models over approved drugs for carcinogenicity, hepatotoxicity, cardiotoxicity, and mutagenicity.

predicted $\text{Log}_{10}(\text{LD50})$ by the model based on intraperitoneal (Model_AT_X) has shown a relatively strong correlation with the experimental $\text{Log}_{10}(\text{LD50})$, with few outliers, shown in the left panel of Fig. 3A. To detect which kinds of compound structures can be most accurately predicted by our model, we have selected the five most accurate compounds regarding the absolute difference between prediction and experimental $\text{Log}_{10}(\text{LD50})$, shown in the right panel of Fig. 3A. Similarly, the performance of intraperitoneal related model AT_Y and the intraperitoneal-related model AT_Z on the rat intraperitoneal data is shown in Fig. 3B and C, respectively; they have achieved strong Pearson correlation of 0.65 and 0.62, respectively. The top five highly accurately predicted compounds are shown in the left panel.

3.3. Performance of binary classification models for carcinogenicity, hERG_cardiotoxicity, mutagenicity, and hepatotoxicity

We built four binary classification models for prediction: carcinogenicity, hERG_cardiotoxicity, mutagenicity, and hepatotoxicity. The data size of the training and test set can be found in Table 1. The performances of those prediction models over epochs are shown in Fig. S1. We can observe that after 500 epochs, most models converged according to these performance metrics over epochs for the test set. The performance of carcinogenicity, hERG_cardiotoxicity, mutagenicity, and hepatotoxicity model at the 2,000 epoch are shown in Table 3.

3.3.1. Model performance of carcinogenicity

Carcinogenicity is a severe side effect. Accurate prediction of whether compounds will lead to carcinogenicity is important to help obtain safer drugs. The performance of carcinogenicity prediction models over epochs in the test set is shown in Fig. S1A. The model has performed well on the test set, with 0.69, 0.66, 0.64, 0.64, and 0.27 for AUC, TPR, precision, accuracy, and MCC at epoch 2,000, respectively (Table S2). Considering the currently available dataset for carcinogenicity has a limited size (~1,000), the model achieved AUC 0.69 is a very promising result. Once more carcinogenicity-related data are available in the future, the model performance must be further greatly improved.

3.3.2. Performance of model predicting hERG_cardiotoxicity

hERG_cardiotoxicity refers to the adverse effects on the heart caused by drugs or substances that inhibit the human Ether-à-go-go-Related Gene (hERG) channel. The hERG_cardiotoxicity is a life-threatening side effect. Many drugs have been associated with hERG_cardiotoxicity; hence, pharmaceutical companies and regulators need to assess the potential for hERG_cardiotoxicity during drug development to ensure patient safety. Accurate prediction of whether compounds will lead to hERG_cardiotoxicity is essential to help obtain safer drugs. The performance of hERG_cardiotoxicity prediction models over epochs in the test set is shown in Fig. S1B. The model has performed very well on the test set, with 0.77, 0.69, 0.68, 0.71, and 0.42 for AUC, TPR, precision, accuracy, and MCC at epoch 2,000, respectively (Table S3). The model accuracy can be further improved as future related data size accumulates.

3.3.3. Performance of model predicting mutagenicity

Mutagenicity refers to the ability of a substance to cause changes (mutations) in the DNA sequence of a cell, which can lead to serious side effects such as abnormal cell behavior and potentially cause genetic disorders or cancer. Experimental evaluation of the mutagenicity of compounds is expensive; hence, accurate prediction of whether compounds by deep learning model is highly desirable and will help obtain safer drugs. The performance of mutagenicity prediction models over epochs in the test set is shown in Fig. S1C. The performance on the test set is very stable, partly because the training and test set of mutagenicity data have the size of 5,500 and 1,376, respectively, much larger than the other three data sets, shown in Table 1. The model has performed very

well on the test set, with 0.88, 0.83, 0.83, 0.81, and 0.62 for AUC, TPR, precision, accuracy, and MCC at epoch 2,000, respectively (Table S4). This result further reinforces our view that the key constraint on the toxicity prediction model performance at this time is the data scale.

3.3.4. Performance of model predicting hepatotoxicity

Hepatotoxicity refers to the toxic or harmful effects of substances or medications on the liver. The liver is a vital organ responsible for metabolizing drugs, detoxifying chemicals, and producing essential proteins. However, certain substances, including medications, alcohol, chemicals, herbal supplements, and viruses, can cause liver damage and dysfunction. The performance of hepatotoxicity prediction models over epochs in the test set is 0.79, 0.73, 0.75, 0.73, and 0.46 for AUC, TPR, precision, accuracy, and MCC at epoch 2,000, respectively, shown in Fig. S1D and Table S5.

3.4. Determine the cutoff value during model application by predicting approved drugs

During application, we need to know which cutoff values are enough to obtain compounds with tolerable toxicity and guarantee that enough candidates are left after toxicity screening. We use our models to predict the approved drug dataset containing 1,768 drugs. We assume most approved drugs (here, we used 80 %) are relatively safe and low toxicity. The predicted value distribution for Acute toxicity is shown in Fig. 4.

Based on the distribution of prediction scores on approved drugs, the threshold values obtained for the acute toxicity prediction models are 2.06 for intraperitoneal administration, 1.62 for intravenous administration, and 2.54 for oral administration, as shown in Fig. 4. Any predicted value lower than these thresholds during the screening application will be considered toxic (a smaller log (LD50) indicates higher toxicity). Similarly, the threshold values for the four binary classifier models are as follows: 0.99, 0.99, 0.77, and 0.75. During screening, any predicted value higher than these thresholds will be discarded. Indeed, the threshold value for hepatotoxicity is relatively high. This is likely because many approved drugs have been predicted to have hepatotoxicity. Consequently, more than relying solely on the binary classification model for hepatotoxicity prediction may be required during real screening applications. Upon reviewing the prediction score distribution from the DrugBank dataset, we observed that even approved drugs can exhibit high scores for carcinogenicity and hepatotoxicity. This observation underscores the challenge in identifying compounds that are completely devoid of such toxic effects. A contributing factor to this issue stems from the relatively low toxicity thresholds used in the training data to classify compounds as positive for carcinogenicity and hepatotoxicity. In essence, compounds with even minimal toxic effects may have been categorized as positive. While the current thresholds may result in a majority of screened compounds being predicted as toxic, they provide a starting point to exclude compounds with substantial risks of carcinogenicity and hepatotoxicity. However, users are encouraged to adjust these thresholds based on their specific requirements. For instance, if a drug's development demands exceptionally low levels of carcinogenicity or hepatotoxicity, lowering the threshold will filter out a greater number of compounds associated with toxicity concerns. As the database continues to expand and more quantitative data on carcinogenicity and hepatotoxicity become available, our predictive capabilities will be further refined. This progress will enable a more accurate and quantifiable assessment of the potential toxicity risks associated with new compounds. Mutagenicity models tend to be relatively small for approved drugs. This suggests that these models can effectively identify and exclude compounds with relevant toxicity from the candidate list. In other words, these models have a higher predictive power in determining whether a compound exhibits cardiotoxic or mutagenic properties, making them valuable tools in the screening process.

Table 4

The compounds candidate list of PDL1 dimer after toxicity screening. The AC1, AC2, AC3, Carcinogenicity (Carc), hERG_cardiotoxicity (hERG), Hepatotoxicity (Hepat) and Mutagenicity (Muta) represent the predicted value of Acute Toxicity (intraperitoneal), Acute Toxicity (intravenous), Acute Toxicity (oral), carcinogenicity, hERG cardiotoxicity, mutagenicity, and hepatotoxicity, respectively. Since carcinogenicity's threshold value is 0.994 and hepatotoxicity is 0.9999941, their predicted values are round to 3 and 7 decimal places, respectively, for clarity, while others round to 2.

Name	AC1	AC2	AC3	Carc	hERG	Hepat	Muta	Group ID
E955-1055	2.92	1.76	3.07	0.685	0.00	0.990612	0.00	1
E955-1623	2.49	1.81	3.08	0.063	0.37	0.999276	0.01	
E955-1627	2.86	1.63	3.08	0.179	0.29	0.007139	0.01	
E955-1769	2.88	1.64	3.05	0.204	0.59	0.115938	0.00	
E955-1274	2.36	1.74	3.02	0.239	0.67	0.947711	0.21	2
2469-0058	3.08	2.07	3.16	0.006	0.01	0.999938	0.04	3
2469-0074	2.99	1.77	3.23	0.000	0.17	0.999922	0.00	
3617-3454	3.07	1.91	3.20	0.000	0.00	0.999992	0.01	
3617-3457	3.09	1.80	3.08	0.000	0.00	0.999981	0.01	
3617-4109	3.06	1.62	3.29	0.000	0.00	0.999799	0.00	
7840-3238	3.02	1.90	3.34	0.008	0.51	0.999993	0.00	
7840-3247	3.02	1.91	3.40	0.004	0.54	0.999848	0.00	
F603-0754	3.05	2.04	3.29	0.001	0.00	0.999938	0.01	
3365-6364	2.90	1.71	3.11	0.001	0.00	0.992061	0.57	4
8014-7009	3.04	1.69	3.01	0.000	0.00	0.986242	0.00	
G715-1358	3.17	1.73	3.27	0.004	0.03	0.996701	0.00	
8290-03341	2.84	1.78	2.89	0.002	0.00	0.999994	0.38	5
D725-0075	2.73	1.75	2.89	0.000	0.67	0.999673	0.01	
2265-3145	2.80	1.91	3.41	0.198	0.00	0.999937	0.00	6
3852-0327	2.23	1.96	3.12	0.000	0.00	0.904507	0.33	7
4151-0291	2.08	2.14	3.11	0.000	0.62	0.998844	0.00	
8003-1716	2.91	1.72	2.61	0.798	0.09	0.001391	0.01	
8010-5343	3.06	1.95	2.81	0.000	0.00	0.289620	0.18	
8629-0047	2.91	1.76	2.92	0.915	0.00	0.992472	0.00	
C487-0210	2.93	1.95	3.15	0.054	0.00	0.999696	0.09	
D321-0703	2.96	2.07	3.19	0.978	0.00	0.970240	0.00	
D656-0104	2.94	1.80	2.86	0.981	0.00	0.910673	0.00	
E667-1007	2.96	2.55	3.17	0.067	0.00	0.999846	0.00	
E894-0882	2.92	1.67	3.08	0.010	0.00	0.999994	0.01	
F431-0440	2.90	2.35	3.01	0.001	0.00	0.998440	0.00	
F506-0281	2.91	1.63	2.98	0.000	0.66	0.999027	0.00	
F506-0852	2.90	1.79	3.01	0.039	0.00	0.999985	0.00	
G808-0713	2.84	1.85	2.96	0.853	0.00	0.999547	0.07	
J021-0016	2.98	2.15	3.07	0.970	0.00	0.002020	0.00	
J021-0098	2.94	2.32	3.09	0.978	0.00	0.000657	0.00	
L118-0057	2.91	2.32	3.33	0.993	0.67	0.998549	0.00	
L279-0439	2.64	2.01	2.92	0.988	0.40	0.003919	0.01	
M460-2207	2.87	1.97	2.92	0.081	0.16	0.999994	0.36	
S721-0237	2.95	1.69	2.97	0.002	0.00	0.013079	0.00	
S763-0925	2.25	2.04	3.17	0.983	0.00	0.974804	0.00	
S951-0223	2.49	2.14	3.31	0.000	0.00	0.999888	0.00	
V004-4306	2.95	1.85	3.10	0.085	0.00	0.999461	0.00	
V004-4354	3.03	1.99	3.18	0.000	0.00	0.998935	0.49	
V022-6200	2.96	1.86	3.07	0.024	0.00	0.999227	0.00	
M769-0707	2.77	2.24	3.13	0.971	0.00	0.997710	0.00	8
M769-0739	2.76	2.40	3.22	0.774	0.00	0.999983	0.00	
M769-0744	2.79	2.36	3.19	0.932	0.00	0.999976	0.00	
P704-0263	2.59	2.73	3.15	0.000	0.00	0.998193	0.00	
P704-0273	2.64	2.49	3.15	0.000	0.00	0.999991	0.00	
F617-0152	2.61	1.99	2.71	0.884	0.01	0.871493	0.00	9
Y043-4448	2.57	1.67	3.50	0.000	0.00	0.002536	0.01	
Y043-4611	2.83	2.32	3.46	0.006	0.00	0.000245	0.00	
C301-3800	2.69	1.91	3.00	0.001	0.01	0.991662	0.00	10
C301-3802	2.78	1.99	2.97	0.023	0.00	0.999810	0.00	
C301-4484	2.63	1.91	3.02	0.001	0.00	0.993771	0.00	
C301-4556	2.73	1.99	2.99	0.049	0.00	0.999875	0.00	

4. Discussion

The approaches of data curation, augmentation, and transfer learning are employed to augment the generalizability of prediction models, especially in situations where data may be constrained or influenced by bias. By utilizing these methodologies in conjunction, scholars can augment the applicability of prediction models and enhance their efficacy over a wide range of datasets and practical scenarios. Nevertheless, it is crucial to meticulously evaluate the distinct attributes of the dataset and task in question while choosing and implementing these techniques to guarantee their efficacy. Furthermore, it is imperative to conduct comprehensive validation and assessment in order to evaluate the effects of these strategies on the performance of the model and its ability to generalize. When incorporated into the screening pipeline (Fig. S3), the toxicity prediction can aid in the early exclusion of potentially dangerous substances. The figure illustrates the integration of deep learning techniques for drug screening, toxicity prediction models, molecular docking, force field-based screening, and experimental validation to identify thousands of chemical compounds and rank them in the top five. To illustrate how to use the developed models in real applications, we use the virtual screening against PDL1 dimer as a concept-of-approve example of toxicity prediction model application. We previously screened compounds for PDL1 dimer [42]. However, we haven't considered the toxicity of compounds during candidate selection. Here, we select 1,327 compounds with DeepBindGCN_BC scores ≥ 0.99 and DeepBindGCN_RG scores ≥ 8.2 values and apply our seven toxicity prediction models to select the candidates with a potential affinity and high safety. The cutoff value is chosen based on Fig. 4's threshold value. Finally, we obtained 56 candidates, shown in Table 4. We observed that many suspected toxicity compounds could be removed. We can make the selected compounds easier to pass the later stage of drug discovery, which significantly reduces the cost of developing new drugs. We clustered these high-score compounds into ten groups based on their similarities by Clusfips (<https://github.com/kaiwang0112006/clusfips>) with the algorithm of Murtagh [43], which depends on RDKit [44]. Other experimental researchers may gain clues from those compounds for developing drugs against PDL1 dimer.

Some types of toxicity prediction still need to be made easier to predict due to limited data size. For instance, Reproductive Toxicity has few datasets, so we still need to train a reliable model to predict it. The GCN can represent the molecule's physical–chemical features and spatial information well. However, there has been more advanced representation recently, which can be tested in the future and may further improve the performance. Our model may implicitly identify the relation of compound features with toxicity. But, like many other methods, it fails to explain explicitly why certain compounds have potential toxicity. Further, there is a known discrepancy between the outcomes of animal model studies and human studies in practice [45]. Although our models have been shown to perform well on the chosen datasets, they may have slightly poorer performance when applied to human-based datasets due to this discrepancy. The species differences between humans and experimental animals significantly challenge drug development [46]. Although experimental animals can be used to assess the safety and efficacy of drugs preliminarily, they cannot fully represent humans' physiological and pathological characteristics. There are differences between humans and other species regarding genetics, metabolism, anatomical structures, physiological functions, etc. Therefore, the effects of drugs on animals may not completely apply to humans [47].

Based on existing scientific and technological means and regulatory frameworks, it is almost impossible to bypass animal testing altogether and directly develop drugs for human use [6]. Animal testing remains the gold standard in non-clinical toxicology research [45]. The toxicological results obtained from animal experiments are essential in guiding clinical trials and are crucial for clinical trial risk control [48]. On the other hand, although we got some training data based on animal models

from the toxic database, obtaining enough human training data remains a significant challenge. We are appealing to the community to release more human-related data to help address this issue. Despite these challenges, the model we developed in the study contributes to predicting relevant toxicity in the early stages of drug development. The current research places significant importance on ensuring the quality and diversity of the dataset utilized. This is essential in order to train models that can accurately capture the fundamental patterns associated with drug toxicity. Furthermore, we investigated various efficient methodologies for data augmentation in order to artificially enhance the dataset, hence enhancing the model's understanding of the fundamental data distribution. Additionally, it is important to acknowledge the significance of transfer learning in harnessing knowledge from interconnected domains or extensive datasets to enhance the efficacy of our models. The most significant performance limitation factor is the dataset size for most toxicity prediction models we tried in this work. But in the future, if the size of publicly available data has significantly increased, or companies/institutes have lots of their unique dataset, the model architecture can also be improved, as suggested in Fig. 1A.

5. Conclusion

Drug toxicity is the most common cause that leads to drug candidate's failure, significantly increasing the drug development cost and risk. Identifying drug toxicity at a later stage would make the whole effect useless. Hence, we develop toxicity prediction models that identify toxicity compounds early in drug screening. We have developed several deep learning-based models to predict several types of drug toxicity, including acute toxicity, carcinogenicity, hERG cardiotoxicity, mutagenicity, and hepatotoxicity. The models perform well in the test set, with Pearson R values 0.76, 0.74, and 0.65 for acute toxicity with intraperitoneal intravenous and oral as routes of administration. Notably, we have used different strategies during the training for different types of available data to achieve reliable results. Furthermore, we have taken as a self-proof of concept example to demonstrate the screening process that integrates our developed toxicity prediction tools. In conclusion, developing toxicity prediction models is crucial for identifying toxic compounds early in drug screening. Our study has demonstrated the effectiveness of deep learning-based models in predicting several types of drug toxicity, with high accuracy rates in the test set. We have also utilized different strategies to train the models based on the available data for each type of toxicity. Additionally, we have illustrated the integrated screening process using a PDL1 dimer, a self-proof of concept example of those toxicity prediction model usage during virtual screening. Through our study, we have provided tools and new ideas for deep learning-based toxicity prediction and shown their potential to facilitate drug development. The construction strategy of the models may provide a foundation for future tool development. Overall, our work has the potential to advance the field of drug development and improve the success rate of drug candidate screening.

Author contributions

HZ designed the study. HZ performed the coding, model construction, and computations, HZ, KMS, JW, LD, JZ, JZHZ performed the data analyses. All authors contributed to writing the manuscript. HZ supervised the study. All authors read and approved the final manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The proposed models and scripts are available upon request from the corresponding author.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ymeth.2024.04.020>.

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